UNIVERSITY OF RUHUNA
DEPARTMENT OF MATHEMATICS

Bachelor of Science (General) Degree (Level I)
MATHEMATICS
MAT 121β : Algebra

Tutorial 04 **Semester II, 2026**

Submit the answer sheet for only 2(a),2(b), 3(a), 3(d), 4(a) and 7 questions on or before: 29/06/2026

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1. If $|A| = -7$, where A is 3×3, find:

a) $|3A|$
b) $|A^{-1}|$
c) $|(2A)^{-1}|$
d) $|2A^{-1}|$
e) $|A^T|$

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2. (a) If $|4A| = 1296$, where A is 3×3, find $|A|$

(b) Suppose $|A| = -6, |B| = 4, |C| = -2$, where A, B and C are 3×3 matrices. Find

(i) $|ABC|$
(ii) $\left|3A^{-1}B^T\right|$
(iii) $\left|(2A)^T B^{-1}\right|$
(iv) $\left|4A^{-1}(2B)^T C\right|$

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3. Without expanding the determinant , evaluate

(a) $\begin{vmatrix} 2 & 5 & 7 \\ 6 & 15 & 21 \\ 4 & 1 & 3 \end{vmatrix}$

(b) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

(c) $\begin{vmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 5 & 7 & 2 & 8 \\ 1 & 6 & 4 & 9 \end{vmatrix}$

(d) $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 5 & 7 \\ 3 & 5 & 8 & 11 \\ 4 & 7 & 11 & 15 \end{vmatrix}$

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4. Use row reduction to show that

(a) $\begin{vmatrix} x+y & y+z & z+x \\ y+z & z+x & x+y \\ z+x & x+y & y+z \end{vmatrix} = -2(x+y+z)\left(x^2+y^2+z^2-xy-yz-zx\right)$

(b) $\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = (b-a)(c-a)(d-a)(c-b)(d-b)(d-c)$

(c) $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = ab+ac+bc+abc$

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5. Find all values of k such that $\begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix} = 0$

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6. Using Cramer's Rule, solve

$$2x - y + z = 6,$$
$$x + 2y - z = 3,$$
$$3x + y + 2z = 11.$$

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7. A University offers three academic programs Medicine, Engineering and Law. The total number of students is 1500. Engineering has 200 more students than Medicine, and Law has 50% of the Engineering students. In addition, due to a new intake policy, the university increases admissions next year by 10% for Medicine, 5% for Engineering, while Law remains unchanged. Determine the number of students in each program after the percentage increase.

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8. Use the matrix inverse method to find the solution set of

$$2x + y - z = 8,$$
$$x + 3y + 2z = 13,$$
$$3x - y + z = 4.$$
